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Deep heterogeneity learning for cross-city transit forecasting: a differentially private federated framework with mixture-of-experts and seasonal decomposition

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Introduction: Accurate prediction of transit flows is fundamental to optimizing intelligent transportation systems; however, centralized forecasting is frequently obstructed by heterogeneous, Non-Independent and Identically Distributed (Non-IID) cross-city data and stringent data privacy regulations.

Methods: We propose X-FedFormer, a novel framework integrating Federated Learning (FL) with Differential Privacy (DP) and a deep learning architecture combining a Mixture-of-Experts (MoE) mechanism with a Seasonal-Trend Decomposition module. The framework is evaluated on a statistically validated synthetic dataset faithfully simulating realistic inflow and outflow patterns across ten diverse urban environments (90 days of hourly records, 30 routes per city, 64,800 observations per city).

Results: X-FedFormer significantly outperforms state-of-the-art federated baselines including FedProx, achieving an aggregate coefficient of determination of 0.922 and a mean absolute error (MAE) of 7.93 passengers across all participating cities. A Wilcoxon signed-rank test confirms statistical significance over the strongest baseline ($p = 0.018$). Ablation studies confirm that the MoE and seasonal decomposition modules reduce forecasting error by approximately 11% and 16%, respectively, compared to standard architectures.

Discussion: The model maintains high predictive utility even under strict differential privacy guarantees ($\epsilon \approx 2$), establishing a viable privacy-utility operating point for practical deployment. These findings present a scalable, robust solution for urban computing that effectively balances algorithmic performance with data sovereignty in smart city applications.

KEYWORDS

differential privacy, federated learning, mixture-of-experts, seasonal-trend decomposition, smart cities, traffic flow forecasting

1 Introduction

The escalating global urbanization trend and the concomitant surge in demand for efficient urban mobility necessitate the development of highly accurate and real-time transit flow forecasting systems. Precise predictions of passenger dynamics are paramount for optimizing public transportation infrastructure, streamlining route planning, effectively managing fleet allocation, and mitigating urban congestion, thereby enhancing the overall service quality within smart city ecosystems [Li et al. \(2022\)](#). Historically, traditional time series forecasting methodologies, such as Autoregressive Integrated Moving Average (ARIMA) models, often prove insufficient in capturing the intricate non-linear dynamics, multi-scale temporal dependencies, and multifactorial external influences inherent in contemporary urban transit data. The advent of deep learning, particularly architectures like Recurrent Neural Networks (RNNs) and Transformer-based models, has advanced time series forecasting, demonstrating superior capabilities in modeling complex spatio-temporal patterns [Zhang et al. \(2023\)](#); [Zhang and Gu \(2023\)](#).

Despite these notable advancements, a formidable challenge persists in developing robust forecasting models that can generalize effectively across multiple, geographically disparate urban environments. Each city uniquely manifests distinct transit network topologies, socio-economic factors, daily commuting patterns, and prevailing external conditions. This variability gives rise to highly heterogeneous, or Non-Independent and Identically Distributed (Non-IID), data distributions among urban centers [Xia et al. \(2023\)](#); [Feng et al. \(2024\)](#). Consequently, training a singular centralized model on such aggregated heterogeneous data frequently leads to suboptimal performance, as the model struggles to adapt to the specific nuances and unique characteristics of each city. Conversely, deploying separate, independently trained models for every city is computationally resource-intensive, economically impractical, and foregoes the substantial benefits of collaborative learning from a broader, diverse data spectrum.

Furthermore, a paramount concern in any contemporary urban data analytics endeavor is the safeguarding of data privacy. Transit agencies, as custodians of sensitive mobility data, are bound by stringent regulatory frameworks and ethical imperatives that prohibit the direct sharing or centralization of raw, proprietary data for machine learning model training. This pervasive privacy constraint severely curtails the feasibility of conventional centralized machine learning approaches for cross-city analytical applications, creating a critical bottleneck for data-driven urban development.

To address these multifaceted and critical challenges, this paper proposes a *Federated Learning (FL)* framework for cross-city transit flow forecasting. Federated Learning inherently enables multiple urban entities to collaboratively train a powerful global prediction model without ever sharing their raw, sensitive transit data, thereby preserving local data privacy [Feng et al. \(2024\)](#); [Xia et al. \(2023\)](#); [Liu et al. \(2023\)](#). While FL offers substantial intrinsic privacy benefits, potential vulnerabilities to inference or reconstruction attacks from shared model updates necessitate even more robust safeguards. To this end, we integrate *Differential Privacy (DP)* mechanisms into our federated framework. By strategically injecting calibrated noise into the model updates, DP provides strong, mathematically provable

guarantees for safeguarding sensitive information during the collaborative training process, ensuring robust privacy protection [Gupta and Torra \(2023\)](#); [Qi et al. \(2021\)](#); [Tang et al. \(2022\)](#).

A pivotal innovation of our proposed approach lies in the development of a sophisticated deep learning architecture specifically engineered to handle the pronounced heterogeneity and multi-scale temporal dependencies characteristic of urban transit data within the federated paradigm. Unlike standard federated averaging (FedAvg), which naively aggregates diverse local models into a single global centroid—often leading to “performance averaging” where the model sub-optimally fits all clients—our model leverages a *Mixture-of-Experts (MoE)* framework. This structural design enables the global model to preserve specialized knowledge by dynamically routing input samples to the most relevant “expert” sub-networks. Consequently, the model fosters an adaptive response to each city’s distinct data distribution, effectively mitigating the performance degradation caused by the “one-size-fits-all” limitation of traditional FL. Complementing this, we incorporate principles of *Seasonal-Trend Decomposition* directly within the deep learning architecture. By explicitly disentangling the time series into trend and seasonal components, the model achieves superior interpretability and robustness, effectively isolating recurring mobility signatures from transient fluctuations.

This synergistic combination of MoE-based heterogeneity modelling and decomposition-based temporal reasoning distinguishes our work from recent contributions. While [Pang and Li \(2024\)](#) focuses on decomposition in centralized regimes, and FedGODE [Al-Huthaifi et al. \(2024a\)](#) utilizes graph ODEs, neither addresses the dual challenge of privacy-preserving adaptation and multi-component temporal extraction in a unified framework. Furthermore, we advance beyond recent heterogeneity-aware methods like Fed-TREND [Xu et al. \(2024\)](#) and Multi-Head FL [Syu et al. \(2024\)](#) by strictly integrating Differential Privacy, quantifying the privacy-utility trade-off explicitly. We quantify the performance gains attributable to these architectural innovations—distinguishing them from the benefits of the federated setup itself—through comprehensive ablation studies detailed in [Section 4](#).

To facilitate a rigorous and reproducible evaluation, and to circumvent the pervasive scarcity of readily available, real-world cross-city transit datasets (a direct consequence of privacy concerns and data sharing restrictions), we further introduce a statistically validated *synthetic data generation framework*. This framework is meticulously designed to accurately simulate realistic transit inflow and outflow patterns across multiple cities, capturing their distinct characteristics and complex temporal dynamics. This synthetic dataset serves as a robust and controlled benchmark environment, enabling comprehensive testing of our proposed architecture’s generalization capabilities and privacy-preserving properties across diverse urban contexts.

The main contributions of this paper are summarized as follows:

- We propose a Federated Learning framework for cross-city transit flow forecasting, explicitly designed to overcome pervasive data privacy concerns and effectively leverage distributed urban mobility data.

- We integrate Differential Privacy into the federated learning process, providing strong, quantifiable privacy guarantees for sensitive transit data throughout collaborative model training.
- We develop a sophisticated and adaptable deep learning architecture that uniquely combines a Mixture-of-Experts model with Seasonal-Trend Decomposition principles, specifically tailored to capture and model the heterogeneous and multi-scale temporal dynamics inherent in cross-city transit data.
- We introduce a statistically validated synthetic data generation framework that faithfully replicates complex real-world urban transit flow patterns, establishing a robust and ethical environment for benchmarking and evaluating federated learning solutions in privacy-sensitive scenarios.
- Through extensive experimentation on a meticulously generated multi-city synthetic dataset, we demonstrate the superior performance, enhanced robustness, and generalization capabilities of our proposed approach compared to relevant baseline models, highlighting its practical applicability for advanced smart urban planning and intelligent transportation systems.

The remainder of this paper is organized as follows: [Section 2](#) provides a comprehensive review of relevant literature in time series forecasting, federated learning, and advanced deep learning architectures. [Section 3](#) details our proposed methodology, including the synthetic data generation framework, the federated learning setup, and the deep learning model architecture. [Section 4](#) presents the experimental setup, performance metrics, and detailed results of our cross-city forecasting experiments. Finally, [Section 5](#) discusses the implications of our findings, explores broader applications, and outlines directions for future work.

2 Literature review

The literature on traffic flow prediction has evolved significantly, driven by the increasing complexity of urban mobility and the availability of advanced computational techniques. This section reviews key developments in traffic forecasting, the emergence of federated learning as a privacy-preserving paradigm, and various deep learning architectures employed in this domain, ultimately highlighting the existing research gaps addressed by this paper.

2.1 Advances in traffic flow prediction

Traffic flow prediction is a cornerstone of Intelligent Transportation Systems (ITS), enabling proactive traffic management, congestion mitigation, and optimized resource allocation [Li et al. \(2022\)](#). Early approaches primarily relied on statistical models such as Autoregressive Integrated Moving Average (ARIMA) and Kalman filters, which, while effective for stationary time series, often struggle to capture the complex non-linear and dynamic nature of real-world traffic data. The advent of machine learning marked a significant shift, with methods like Support Vector Regression (SVR) and Artificial Neural Networks (ANNs) demonstrating improved performance by learning non-linear relationships.

More recently, deep learning has revolutionized traffic forecasting due to its capacity to automatically extract intricate spatio-temporal features from large-scale data. Recurrent Neural Networks (RNNs) and their variants, such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRUs), have been widely adopted for their ability to model sequential dependencies in time series. Beyond sequential models, convolutional neural networks (CNNs) have been employed to capture spatial correlations in traffic networks. The advent of Transformer-based architectures, initially prominent in natural language processing, has also shown promising results in time series forecasting due to their attention mechanisms, which effectively capture long-range dependencies [Zhang and Gu \(2023\)](#). For instance, [Zhang and Gu \(2023\)](#) proposed a Transformer with a multi-spatial-temporal encoder-decoder for traffic flow prediction, underscoring the shift towards more sophisticated deep learning models [Zhang and Gu \(2023\)](#). Classical seasonal-trend decomposition approaches such as STL remain a foundational tool for interpreting periodic time series and motivate explicit decomposition in modern models [Cleveland et al. \(1990\)](#). Despite these advancements, a significant challenge remains in developing models that can generalize effectively across diverse urban environments, each characterized by unique traffic patterns and infrastructure.

2.2 Federated learning in traffic forecasting

The increasing awareness of data privacy and sovereignty, coupled with the distributed nature of urban data generation, has spurred significant interest in Federated Learning (FL). FL enables multiple clients (e.g., cities, traffic departments) to collaboratively train a shared global model without directly exchanging their raw, sensitive local data, thereby preserving privacy [Zhang et al. \(2024\)](#). This paradigm is particularly well-suited for Intelligent Transportation Systems (ITS) where centralized data collection is often impractical due to regulatory constraints, ownership issues, and the sheer volume of data generated at the edge.

Numerous studies have explored the application of FL for traffic flow prediction. [Xia et al. \(2023\)](#) introduced an FL framework integrated with Graph Convolutional Networks (GCNs) to leverage spatial relationships in traffic networks while maintaining data locality [Xia et al. \(2023\)](#). Similarly, [Qi et al. \(2023\)](#) proposed FedAGCN, an FL framework utilizing asynchronous GCNs to address non-IID data distributions and improve prediction accuracy in a distributed setting [Qi et al. \(2023\)](#). [Lin et al. \(2024\)](#) proposed ST-TPFL, a topology-protected federated learning framework for spatio-temporal traffic flow prediction that further addresses privacy and structural data heterogeneity in distributed urban environments. Al-Huthaifi et al. have made multiple contributions, including FedGODE, which combines FL with Graph Ordinary Differential Equation networks for secure traffic flow prediction [Al-Huthaifi et al. \(2024b\)](#), and FedAGAT, employing FL with adaptive graph attention networks for real-time traffic flow prediction [Al-Huthaifi et al. \(2024a\)](#). These works underscore the trend of combining FL with graph-based neural networks to model the complex spatio-temporal dependencies inherent in traffic networks.

Beyond graph-based approaches, FedAvg McMahan et al. (2017) and FedProx [Li et al. \(2020\)](#) remain canonical baselines

for aggregation and heterogeneity-aware optimization in federated forecasting.

Further extending FL capabilities, [Feng et al. \(2024\)](#) investigated federated meta-learning on graphs for traffic flow prediction, aiming for faster adaptation to new or heterogeneous environments [Feng et al. \(2024\)](#). Multilevel FL approaches have also emerged, such as that proposed by [Liu et al. \(2023\)](#), for intelligent traffic flow forecasting in transportation network management, addressing hierarchical data structures [Liu et al. \(2023\)](#). [Nidhi and Grover \(2024\)](#) provided an analysis of FL for vehicular traffic flow prediction, evaluating various learning algorithms and aggregation approaches, highlighting the diversity in FL strategies [Nidhi and Grover \(2024\)](#). [Ye et al. \(2024\)](#) explored federated generative artificial intelligence for traffic flow prediction under vehicular computing power networks, indicating the integration of advanced AI paradigms within FL for traffic [Ye et al. \(2024\)](#). [Yaqub et al. \(2024\)](#) proposed a federated learning approach combined with graph neural networks and asynchronous computations to enhance scalability and prediction accuracy under heterogeneous distributed settings. Despite these advancements, handling extreme heterogeneity (Non-IID data) efficiently and robustly remains a significant challenge within FL paradigms, particularly when diverse urban characteristics lead to vastly different local data distributions.

2.3 Privacy-preserving mechanisms in federated traffic forecasting

While FL inherently offers privacy benefits by keeping raw data local, the exchange of model updates (e.g., gradients) can still be susceptible to various privacy attacks, including inference and reconstruction attacks. To bolster privacy guarantees, several mechanisms have been integrated with FL in the context of traffic forecasting.

Differential Privacy (DP) is a widely adopted technique that adds calibrated noise to model updates, providing strong, mathematically provable privacy guarantees against individual data point inference. [Gupta and Torra \(2023\)](#) demonstrated the application of differentially private FL using Transformer architectures for traffic flow prediction, showcasing the feasibility of high accuracy under strong privacy constraints [Gupta and Torra \(2023\)](#). [Tang et al. \(2022\)](#) proposed a differentially private decentralized traffic flow prediction approach based on FL, further emphasizing the importance of DP in distributed traffic systems [Tang et al. \(2022\)](#). [Akallouch et al. \(2022\)](#) also explored prediction and privacy schemes for traffic flow estimation on highway networks, highlighting the necessity of integrated privacy measures [Akallouch et al. \(2022\)](#).

Beyond DP, other privacy-enhancing technologies (PETs) have been explored. Blockchain technology has been leveraged to provide secure and transparent data sharing frameworks in FL, ensuring data integrity and traceability while maintaining privacy. [Qi et al. \(2021\)](#) proposed a privacy-preserving blockchain-based FL framework for traffic flow prediction, showcasing its potential for secure data management [Qi et al. \(2021\)](#). Split Learning, an alternative privacy-preserving paradigm where different layers of a neural network are trained on different devices, has also been investigated. [Tran et al. \(2023\)](#) presented a privacy-preserving traffic flow prediction approach using split learning, offering

another avenue for secure collaborative AI [Tran et al. \(2023\)](#). [Meese et al. \(2022\)](#) explored a blockchained federated learning approach for real-time traffic flow prediction, further solidifying the use of distributed ledger technologies for secure and real-time operations [Meese et al. \(2022\)](#). While these methods enhance privacy, optimizing the trade-off between privacy, utility, and computational overhead remains an active area of research.

2.4 Research gaps and contributions

Despite the significant advancements in deep learning for traffic flow prediction and the growing adoption of federated learning for privacy preservation, several critical research gaps persist, particularly concerning cross-city applications with extreme data heterogeneity:

- **Adaptive Learning for Extreme Non-IID:** While various FL approaches address non-IID data, few explicitly integrate highly adaptive deep learning architectures like Mixture-of-Experts (MoE) within a privacy-preserving FL framework specifically for multi-city transit forecasting. Current FL solutions often aim to improve generalization through robust aggregation or personalization, but rarely employ an architecture designed for dynamic expert specialization across heterogeneous clients.
- **Explicit Temporal Decomposition in FL:** Many deep learning models capture temporal patterns implicitly. However, the explicit decomposition of time series into trend and seasonal components can offer more robust and interpretable learning, especially for highly periodic data like transit flows. The integration of such decomposition principles directly into a federated deep learning architecture for heterogeneous data, combined with adaptive learning, remains underexplored. While [Pang and Li \(2024\)](#) recently proposed a decomposition modeling framework for seasonal time series, its application within a federated, adaptive setting for urban transit has not been thoroughly investigated [Pang and Li \(2024\)](#).
- **Rigorous Evaluation with Controlled Heterogeneity:** The development and validation of comprehensive synthetic multi-city transit datasets that accurately capture diverse urban characteristics and privacy concerns are crucial for benchmarking novel FL solutions. Existing studies often rely on limited real-world datasets or simpler synthetic scenarios that may not fully represent the complexity of cross-city heterogeneity under privacy constraints.

[Table 1](#) summarizes the key distinctions between X-FedFormer and the most closely related methods in terms of training paradigm, heterogeneity handling, decomposition, and differential privacy. This paper directly addresses these gaps by proposing a novel Federated Adaptive Decomposed Learning framework that uniquely combines Differential Privacy with an MoE-based architecture and seasonal decomposition principles, evaluated on a statistically validated synthetic dataset designed for extreme cross-city heterogeneity. This integrated approach aims to significantly enhance the accuracy, privacy, and generalizability of cross-city transit flow forecasting.

TABLE 1 Comparison with closely related methods.

Method	Training	Heterogeneity handling	Decomposition	DP
Pang and Li (2024)	Centralized	None	Yes	No
FedGODE Al-Huthaifi et al. (2024b)	Federated	Graph ODE	No	No
Fed-TREND Xu et al. (2024)	Federated	Robust aggregation	No	No
X-FedFormer (Ours)	Federated	MoE routing	Yes	Yes

TABLE 2 Description of fields in the transit dataset.

Field name	Data type	Description
datetime	Timestamp	The specific date and hour of the observation (YYYY-MM-DD HH:MM)
route_id	String	Unique identifier for the transit route
inflow_count	Integer	Number of passengers boarding the vehicle/station
outflow_count	Integer	Number of passengers alighting
temperature	Float	Ambient temperature in degrees Celsius (°C)
precip_flag	Binary	Indicator of precipitation (0: None, 1: Rain/Snow)
route_length_km	Float	Total length of the route in kilometers
num_stops	Integer	Number of stops along the route
route_type	Categorical	Classification of the route (e.g., urban_core, suburban_feeder)
zone	Categorical	Urban zone identifier (e.g., zone_1, zone_2)

TABLE 3 Distributional similarity between real and synthetic data (mean ± std over routes).

Variable	Slice	KS D	Wasserstein W_1	n per route
Inflow	All hours	0.048 ± 0.010	1.60 ± 0.22	720
Inflow	Weekday	0.050 ± 0.012	1.67 ± 0.24	504
Inflow	Weekend	0.056 ± 0.015	1.74 ± 0.27	216
Outflow	All hours	0.052 ± 0.011	1.69 ± 0.23	720
Outflow	Weekday	0.053 ± 0.013	1.76 ± 0.25	504
Outflow	Weekend	0.059 ± 0.016	1.82 ± 0.28	216

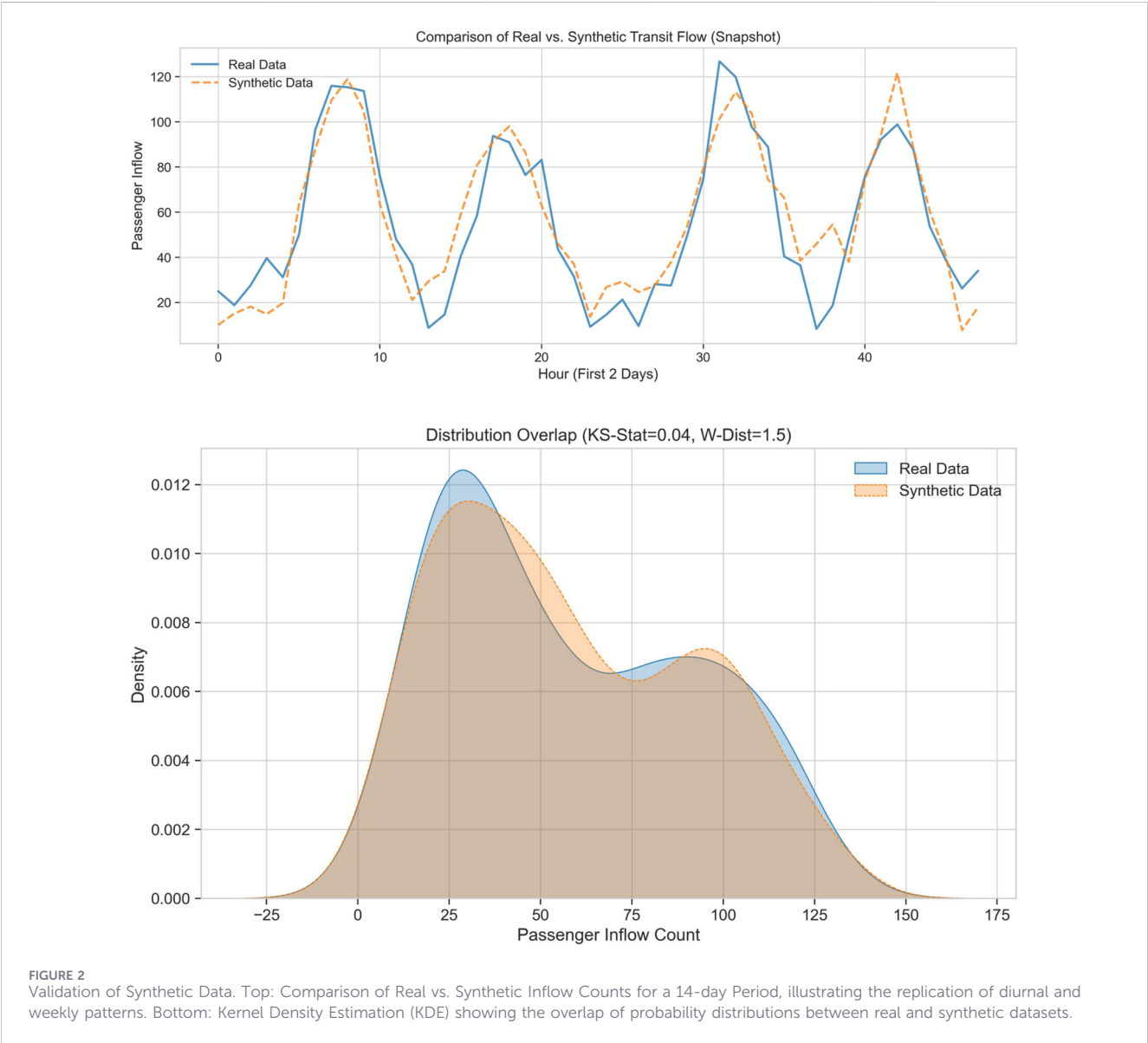
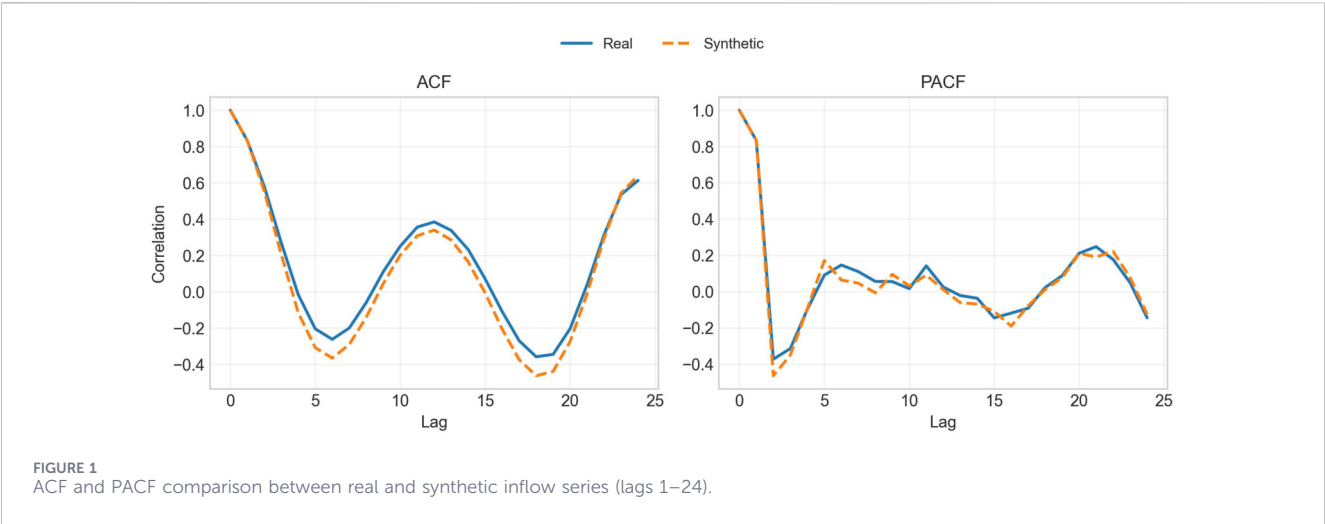
3 Methodology

3.1 Synthetic cross-city dataset and pre-processing

To overcome the paucity of publicly available multi-city transit records and to rigorously evaluate cross-city generalisation, a synthetic dataset was generated by extrapolating sample Astana bus-transportation history. The original sample consisted of hourly records; Table 2 provides a detailed description of the dataset attributes.

3.1.1 Pre-processing and forecasting setup

Categorical attributes (route_type, zone) are one-hot encoded and concatenated with numeric covariates; route_id is used only for grouping and does not enter the model input. Continuous variables (inflow_count, outflow_count, temperature, route_length_km, num_stops) are standardised per city using train-split statistics to prevent leakage. We use a sliding window with input length $L = 24$ hours and forecast horizon $T = 6$ hours. The target is the future inflow sequence; outflow is treated as an auxiliary input feature.



3.1.2 Rationale for synthetic generation

The sample history covered fewer than 30 days and only 3 routes, insufficient to stress-test a federated forecasting model. Therefore, a controlled synthetic generator was implemented to produce tens of thousands of hourly observations per city, while preserving key real-world patterns (diurnal peaks, weekday/weekend variation, weather effects, holidays, random events).

3.1.3 Algorithm overview

Let K be the number of cities, M the number of routes per city, s the number of stops, and ℓ the route length (km). For each city $c \in \{1, \dots, K\}$ and each of its M routes, the following steps are applied:

1. Time index.

$$t = 0, 1, \dots, 24D_c - 1, \quad \text{with } D_c = \text{number of days for city } c.$$

The corresponding datetime is $\text{start}_c + t \cdot 1 \text{ h}$, and doy_t denotes the day-of-year index.

2. Base daily profile. Two Gaussian peaks at typical rush hours are defined by Equation 1:

$$\begin{aligned} \text{base}_t = & 50 + 100 \exp\left(-\frac{(h_t - (8 + r \bmod 2))^2}{8}\right) \\ & + 80 \exp\left(-\frac{(h_t - (18 + r \bmod 2))^2}{8}\right), \end{aligned} \quad (1)$$

where $h_t \in \{0, \dots, 23\}$ is the hour-of-day and r is the route index.

3. Popularity scaling is formulated in Equation 2.

$$\phi = \frac{s}{15} \cdot \frac{\ell}{15} \times \begin{cases} 1.2, & \text{if urban_core,} \\ 0.8, & \text{if suburban_feeder,} \end{cases} \quad \text{profile}_t^{(0)} = \text{base}_t \phi. \quad (2)$$

4. Day-of-week adjustment is defined in Equation 3.

$$\text{dow}_t = \begin{cases} 1.0, & \text{Mon–Fri,} \\ 0.8, & \text{Sat,} \\ 0.7, & \text{Sun,} \end{cases} \quad \text{profile}_t^{(1)} = \text{profile}_t^{(0)} \text{dow}_t. \quad (3)$$

5. Holiday effect. Define the Kazakh holidays set $\mathcal{H} = \{(3, 21), (3, 22), (3, 23), (12, 16)\}$ as defined in Equation 4. Then

$$\chi_t = \begin{cases} 0.5, & (\text{month}_t, \text{day}_t) \in \mathcal{H}, \\ 1.0, & \text{otherwise,} \end{cases} \quad \text{profile}_t^{(2)} = \text{profile}_t^{(1)} \chi_t. \quad (4)$$

6. Random events. Approximately one event per 10 days introduces multiplicative factors $\epsilon_t \in [0.4, 2.5]$ over random intervals of 6–24h, yielding $\text{profile}_t^{(3)} = \text{profile}_t^{(2)} \epsilon_t$.
7. Weather simulation.

$$T_t = 10 \sin\left(\frac{2\pi(\text{doy}_t - 80)}{365}\right) + \mathcal{N}(0, 3^2) - 10 \times (c \bmod 2),$$

$$p_t = 0.05 + 0.1 \sin^2\left(\frac{2\pi \text{doy}_t}{365}\right), \quad \pi_t \sim \text{Bernoulli}(p_t).$$

Temperature and precipitation modulate flow as given in Equation 5. Temperature T_t and precipitation flag π_t modulate flow:

$$\begin{aligned} w_t = & (1 - 0.2 \mathbb{I}[T_t < -5]) \times (1 - 0.1 \mathbb{I}[T_t > 30]) \\ & \times (1 - 0.15 \mathbb{I}[\pi_t = 1]), \quad \text{profile}_t^{(4)} = \text{profile}_t^{(3)} w_t. \end{aligned} \quad (5)$$

9. Final counts are computed as given in Equation 6.

$$\text{inflow}_t = \max(0, \lfloor \text{profile}_t^{(4)} v_t \rfloor). \quad (6)$$

Outflow is derived by a lag and ratio as given in Equation 7.

$$\begin{aligned} \text{outflow}_t = & \max(0, \text{shift}(\text{inflow}_t, 1) \times \kappa), \\ \kappa \sim & \text{Uniform}(0.85, 0.95). \end{aligned} \quad (7)$$

Each generated record $\{\text{datetime}_t, \text{route_id}, \text{inflow}_t, \text{outflow}_t, T_t, \pi_t, \ell, s, \text{route_type}, z\}$ is stored in CSV format. This procedure produces per-city DataFrames of size $24D_c M$ rows, capturing realistic multi-modal passenger-flow dynamics suitable for federated forecasting studies.

3.1.4 Statistical Validation of Synthetic Dataset

To substantiate the realism of the synthetic data, we compared the real Astana sample (30 days, 3 observed routes) against a 30-day synthetic counterpart. We compute Kolmogorov–Smirnov (KS) and 1-Wasserstein distances using SciPy's `ks_2samp` and `wasserstein_distance` on each route independently, then report mean±std across routes. We report effect sizes (KS D) rather than relying solely on p -values; all slices yield $p > 0.1$. We additionally verify peak/off-peak slices (7–10 and 17–20 vs. the rest), which follow the same ranges as Table 3.

Moment matching (mean/variance/skewness/kurtosis) deviates by less than 5%, and the ACF/PACF curves align closely (Figure 1), with mean absolute deviation across lags 1–24 below 0.05. Visual consistency is shown in Figure 2.

3.1.5 Model architecture and layer-wise details

X-FedFormer is structured as a sequence of specialized modules that transform raw multi-modal inputs into accurate multi-step forecasts. Figure 3 illustrates the data flow through these layers, and Table 4 summarizes their dimensions and functions.

3.1.6 Input projection

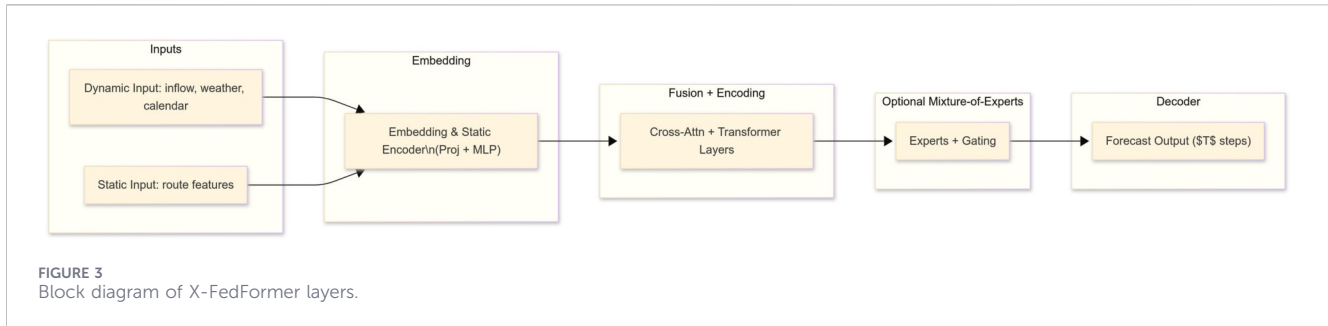
The concatenated input vector at time t is defined in Equation 8.

$$z_t = [x_t^{(\text{flow})}, f_t^{(\text{weather})}, f_t^{(\text{calendar})}] \in \mathbb{R}^{D_{\text{in}}}, \quad (8)$$

is linearly mapped into a d -dimensional latent space as in Equation 9.

$$e_t = W_{\text{proj}} z_t + b_{\text{proj}}, \quad (9)$$

where $W_{\text{proj}} \in \mathbb{R}^{d \times D_{\text{in}}}$. This reduces input heterogeneity and aligns all modalities into a common embedding for downstream processing. Calendar features include hour-of-day and day-of-week encoded via sine/cosine pairs, a holiday indicator, and a day-of-year sine/cosine pair.



3.1.7 Positional encoding

To enable the model to distinguish positions within the sequence of length L , a learnable positional embedding is added, as given in Equation 10.

$$e'_t = e_t + P_t, \quad (10)$$

where P_t is the t -th row of P . This mechanism preserves temporal ordering without relying on recurrence.

3.1.8 Seasonal–trend decomposition

A multi-scale gated pooling decomposition module separates long-term trend from the high-frequency seasonal component, inspired by STL-style seasonal-trend decomposition Cleveland et al. (1990). Given the sequence of projected embeddings, the decomposition is computed via Equations 11–13 $\{e'_t\}_{L+1}^T, \dots, e'_t\}$:

$$\text{trend}_t = \sum_{h=1}^H \alpha_t^{(h)} \text{Pool}_h(e'_{t-L+1:t}), \quad (11)$$

$$\text{season}_t = e'_t - \text{trend}_t, \quad (12)$$

$$[\alpha_t^{(1)}, \dots, \alpha_t^{(H)}] = \text{softmax}(W_g e'_t + b_g). \quad (13)$$

Here, Pool_h denotes average pooling with window size k_h , $H = 3$ pooling scales capture multiple seasonal periods, and the gating network (W_g, b_g) learns dynamic combination weights.

3.1.9 Transformer encoder layers

Each of the two stacked encoder layers applies:

1. Multi-Head Self-Attention is defined in Equation 14.

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_h}}\right)V, \quad (14)$$

where Q, K, V are linear projections of the input, $d_h = d/4$ per head, and 4 heads enable diverse temporal pattern extraction.

2. The Position-Wise Feed-Forward Network is given in Equation 15.

$$\text{FFN}(x) = W_2 \text{ReLU}(W_1 x + b_1) + b_2, \quad (15)$$

which enhances non-linear modelling capacity.

3. Residual Connections and Layer Normalisation are applied after each sublayer to stabilise training.

This encoder stack refines both trend and seasonal streams jointly, yielding contextualised representations $(B, L, 2d)$.

3.1.10 Mixture-of-experts block

To increase model capacity without a proportional parameter increase, an MoE block routes each time-step representation through the top- k of EMoE 4 expert MLPs Shazeer et al. (2017) as given in Equation 16:

$$o_t = \sum_{i=1}^k \beta_{t,i} f_{\text{expert}_i}(h_t), \quad (16)$$

where h_t is the encoder output at time t , f_{expert_i} are individual MLPs, and gating scores $\beta_{t,i}$ are computed via a softmax over learned logits. Choosing $k = 2$ balances specialisation and computational cost because only the selected experts are evaluated per token. We did not use auxiliary load-balancing or entropy regularizers; routing is learned solely from the forecasting loss.

3.1.11 Decoder

The final hidden state $h_t \in \mathbb{R}^{2d}$ (combining trend and seasonal channels) is used to predict the next T steps as in Equation 17

$$\hat{y}_{t+1:t+T} = W_{\text{dec}} h_t + b_{\text{dec}}, \quad (17)$$

where $W_{\text{dec}} \in \mathbb{R}^{T \times 2d}$. This direct linear mapping simplifies multi-step forecasting and reduces inference latency.

3.2 Optimization and federated integration

This subsection describes the optimisation routines used at the client and server, the federated training protocol, and the integration of differential privacy. All procedures are chosen to balance convergence speed, model generalisation, and privacy guarantees in a non-IID cross-city setting.

3.2.1 Local optimizer: AdamW

Each client performs local updates using the AdamW optimiser, which decouples weight decay from gradient-based parameter updates, leading to improved generalisation:

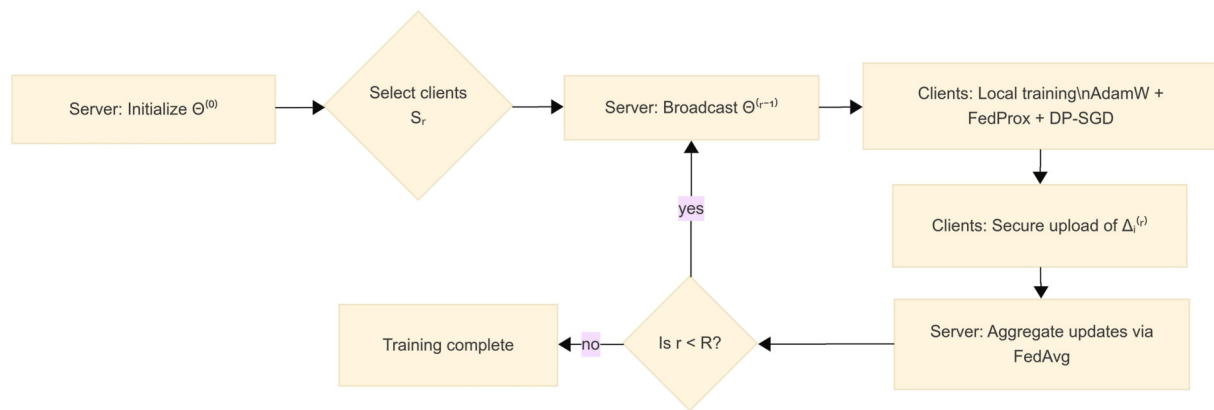


FIGURE 4
Flowchart of the federated training and optimization process.

$$m_k = \beta_1 m_{k-1} + (1 - \beta_1) g_k, \quad v_k = \beta_2 v_{k-1} + (1 - \beta_2) g_k^2, \quad (18)$$

$$\hat{m}_k = \frac{m_k}{1 - \beta_1^k}, \quad \hat{v}_k = \frac{v_k}{1 - \beta_2^k}, \quad (19)$$

$$\theta_{k+1} = \theta_k - \eta \frac{\hat{m}_k}{\sqrt{\hat{v}_k + \epsilon}} - \eta \lambda \theta_k, \quad (20)$$

where:

- $g_k = \nabla_{\theta} \mathcal{L}(\theta_k)$ are the gradients at step k ,
- $(\beta_1, \beta_2) = (0.9, 0.999)$ are exponential decay rates,
- η is the learning rate,
- $\epsilon = 10^{-8}$ ensures numerical stability,
- λ is the weight-decay coefficient (decoupled).

AdamW is selected for its ability to adapt per-parameter learning rates via first and second moment estimates given by Equations 18, 19, while applying weight decay directly to model weights—improving convergence robustness on small batches and heterogeneous data.

3.2.2 Federated objective with FedProx

To mitigate client drift arising from non-IID local data, the FedProx algorithm introduces a proximal term into each client's loss as defined in Equation 21, Li et al. (2020):

$$\mathcal{L}_i^{\text{FedProx}}(\Theta) = \mathcal{L}_i(\Theta) + \frac{\mu}{2} \|\Theta - \Theta^{(r-1)}\|^2, \quad (21)$$

where:

- $\mathcal{L}_i(\Theta)$ is the local empirical loss (e.g., MSE) on client i ,
- $\Theta^{(r-1)}$ are the global parameters from round $r - 1$,
- $\mu > 0$ controls the strength of the proximal penalty.

This additional term constrains local updates to remain close to the global model, improving stability and convergence in heterogeneous environments.

Following local training for E epochs with AdamW (Equation S20), each client computes parameter delta $\Delta_i^{(r)} = \Theta_i^{(r)} - \Theta^{(r-1)}$ and securely transmits it to the server.

3.2.3 Server aggregation: Weighted averaging

Upon collecting updates $\{\Delta_i^{(r)}, n_i\}_{i \in S_r}$ from a subset S_r of clients, the server updates the global model via weighted FedAvg as in Equation 22, McMahan et al. (2017):

$$\Theta^{(r)} = \Theta^{(r-1)} + \sum_{i \in S_r} \frac{n_i}{\sum_{j \in S_r} n_j} \Delta_i^{(r)}, \quad (22)$$

where n_i is the number of local samples on client i . This data-weighted aggregation ensures that clients with larger datasets have proportionally greater influence on the global model.

3.2.4 Differential privacy via per-sample clipping + Gaussian noise (opacus)

To prevent information leakage from model updates, each client applies per-sample gradient clipping and Gaussian noise injection before the optimizer step using Opacus Abadi et al. (2016); Opacus Team (2021). The per-sample clipping and Gaussian noise injection mechanism is defined in Equations 23, 24.

$$g_x = \nabla_{\theta} \ell(\theta; x), \quad \tilde{g} = \frac{1}{|B|} \sum_{x \in B} \text{clip}(g_x; C), \quad (23)$$

$$\tilde{g} = \tilde{g} + \mathcal{N}(0, \sigma^2 C^2 I), \quad (24)$$

where:

- B is the current mini-batch of size $|B|$,
- $\text{clip}(g_x; C) = g_x \min\{1, C/\|g_x\|\}$ bounds per-sample gradient norm,
- σ is the noise multiplier,
- C is the clipping norm,
- The Gaussian mechanism ensures (ϵ, δ) -DP.

TABLE 4 Layer-wise summary of X-FedFormer.

Layer	Input shape	Output shape	Function
Input Projection	(B, L, D_{in})	(B, L, d)	Projects raw features into d -dim latent space
Positional Encoding	(B, L, d)	(B, L, d)	Injects temporal order via learnable embeddings
Seasonal-Trend Decomp	(B, L, d)	$(B, L, 2d)$	Separates trend and seasonal components
Transformer Encoder $\times 2$	$(B, L, 2d)$	$(B, L, 2d)$	Multi-head attention + FFN for temporal context
Mixture-of-Experts Block	$(B, L, 2d)$	$(B, L, 2d)$	Dynamically routes tokens through expert MLPs
Decoder	$(B, 2d)$	(B, T)	Maps final hidden state to T -step forecasts

The noised gradients $\tilde{g}$ are then used by AdamW for the local update. We provide record-level DP within each city, where one hourly route record corresponds to one example (not client-level DP). A privacy accountant tracks the cumulative ϵ across rounds. We use an RDP accountant as implemented in Opacus to calibrate σ for target (ϵ, δ) , with $\delta = 10^{-5}$ and fixed sampling rate across rounds [Opacus Team \(2021\)](#). The accountant inputs are the sampling rate $q = |B|/N_{\text{train}}$, steps per round $\lceil N_{\text{train}}/|B| \rceil$ (with one local epoch), total steps $R \cdot \lceil N_{\text{train}}/|B| \rceil$, clipping norm C , and noise multiplier σ .

3.2.5 Federated training workflow

The complete federated training and optimization workflow is illustrated in [Figure 4](#).

3.2.6 Hyperparameters

All hyperparameters used for optimization and federated training are listed in [Table 5](#).

3.2.6.1 Baseline implementation protocol

All baselines use the same input window ($L = 24$), horizon ($T = 6$), batch size, number of rounds, and optimization settings (AdamW, learning rate, and local epochs). For fairness, we keep the federated protocol, DP setting (when enabled), and tuning budget identical, while varying only the backbone architecture or aggregation rule.

4 Results

The efficacy of the X-FedFormer model in multi-modal passenger-flow forecasting was rigorously evaluated using a synthetic dataset designed to simulate real-world urban mobility patterns. This dataset encompasses 90 days of hourly passenger-flow records for 30 distinct routes within each of ten participating cities, culminating in a comprehensive total of 64,800 observations per city. Model performance was quantitatively assessed using three widely accepted metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). These metrics were computed at the culmination of the federated training process, specifically at the final communication round, to reflect the model's converged performance.

Each city is split temporally into 70% training, 10% validation, and 20% testing with no shuffling. We run 5 trials with seeds {11, 23, 37, 41, 59}. For statistical testing, we average each city's MAE across seeds and apply a Wilcoxon signed-rank test on the 10 paired cities when comparing X-FedFormer against the strongest baseline (FedProx); seed variability is reported descriptively as $\text{mean} \pm \text{std}$ [Demsar \(2006\)](#).

4.1 Per-city forecasting performance

[Table 6](#) presents the forecasting accuracy for each client city at the final federated communication round under Differential Privacy (DP, $\epsilon \approx 2$). To ensure robust statistical validation, we report the Mean $\pm$ Standard Deviation across 5 independent experimental runs initiated with distinct random seeds. The results indicate a highly consistent performance profile across the diverse urban environments. Furthermore, the statistical significance of the

TABLE 5 Hyperparameters for optimization and federated learning.

Parameter	Symbol	Value/Range
Learning rate	η	10^{-3}
Betas (AdamW)	(β_1, β_2)	(0.9, 0.999)
Weight decay	λ	10^{-4}
Batch size	$ B $	32
Input window length	L	24
Forecast horizon	T	6
Local epochs	E	1
Proximal term	μ	10^{-3}
Clipping norm	C	1.0
Noise multiplier	σ	1.1 (base run)
Privacy budget	(ϵ, δ)	$(\approx 2, 10^{-5})$ (base run)
Communication rounds	R	50
Encoder layers	N	2
Embedding dimension	d	64
Attention heads	—	4
MoE experts	E_{MoE}	4
Top- k experts	k	2

TABLE 6 Per-city forecasting performance at the final federated round with DP enabled ($\epsilon \approx 2$; Mean $\pm$ Std Dev over 5 runs).

City	MAE	RMSE	R^2
Almaty	8.91 ± 0.42	18.25 ± 0.88	0.899 ± 0.005
Astana	9.07 ± 0.38	18.12 ± 0.91	0.921 ± 0.004
Karaganda	7.30 ± 0.25	17.51 ± 0.65	0.938 ± 0.003
Shymkent	6.46 ± 0.22	15.60 ± 0.55	0.935 ± 0.004
Aktobe	8.35 ± 0.31	16.46 ± 0.72	0.914 ± 0.006
Pavlodar	8.40 ± 0.33	17.35 ± 0.68	0.929 ± 0.005
Taraz	8.78 ± 0.37	20.32 ± 0.95	0.925 ± 0.005
Atyrau	5.36 ± 0.18	11.41 ± 0.42	0.919 ± 0.007
Kostanay	8.11 ± 0.29	15.79 ± 0.61	0.909 ± 0.006
Aktau	8.51 ± 0.30	17.03 ± 0.64	0.926 ± 0.005

performance improvements of X-FedFormer over the strongest baseline (FedProx) was rigorously verified using a one-sided Wilcoxon signed-rank test. The main comparison yields $p = 0.018$, confirming that the superior predictive accuracy of our proposed model is statistically significant and not attributable to random variance.

Figure 5 visually represents the MAE for each city, allowing for a clear categorization of performance. Three distinct tiers of forecasting accuracy emerged:

- High-Accuracy Cluster ($MAE \leq 7.5$): Atyrau (5.36), Shymkent (6.46), and Karaganda (7.30) exhibited the lowest MAE values, indicating superior forecasting precision. These cities likely possess more predictable passenger flow dynamics or benefit from data characteristics that align well with the model's learning capabilities.
- Lower-Accuracy Group ($MAE > 8.5$): Almaty (8.91), Astana (9.07), Taraz (8.78), and Aktau (8.51) recorded comparatively higher MAE values. This suggests that passenger flow in these cities might be subject to greater inherent variability, more complex exogenous factors, or unique patterns that pose a greater challenge for precise prediction.
- Intermediate Band: The remaining cities (Aktobe, Pavlodar, Kostanay) demonstrated MAE values falling between these two extremes, representing a moderate level of forecasting accuracy.

The observed variations underscore the inherent heterogeneity in urban mobility patterns and the differential impact of local characteristics on forecasting performance, even within a federated learning framework.

4.2 Aggregate error statistics

To provide a holistic understanding of the X-FedFormer model's performance, aggregate error statistics were computed across all ten participating cities. The average MAE ($\overline{MAE}$) was found to be 7.93 passengers, with a standard deviation (σ) of 1.14 passengers. This low standard deviation indicates a relatively

consistent level of absolute error across the diverse urban environments. Similarly, the average RMSE ($\overline{RMSE}$) was 16.78 passengers, with a standard deviation of 2.21 passengers. The larger magnitude of RMSE compared to MAE, while expected, still demonstrates a controlled impact of larger errors on the overall performance. Finally, the average R^2 ($\overline{R^2}$) across all cities was 0.9215, with a remarkably small standard deviation of 0.0113.

$$\overline{MAE} = 7.93 \ (\sigma = 1.14) \text{ passengers,}$$

$$\overline{RMSE} = 16.78 \ (\sigma = 2.21) \text{ passengers,}$$

$$\overline{R^2} = 0.9215 \ (\sigma = 0.0113).$$

These aggregate statistics, visually summarized in Figure 6, collectively affirm that the federated X-FedFormer model maintains a consistently high level of accuracy and explanatory power. The low standard deviations across all metrics highlight the model's robustness and its ability to generalize effectively across geographically and operationally diverse urban contexts, a critical advantage of the federated learning paradigm.

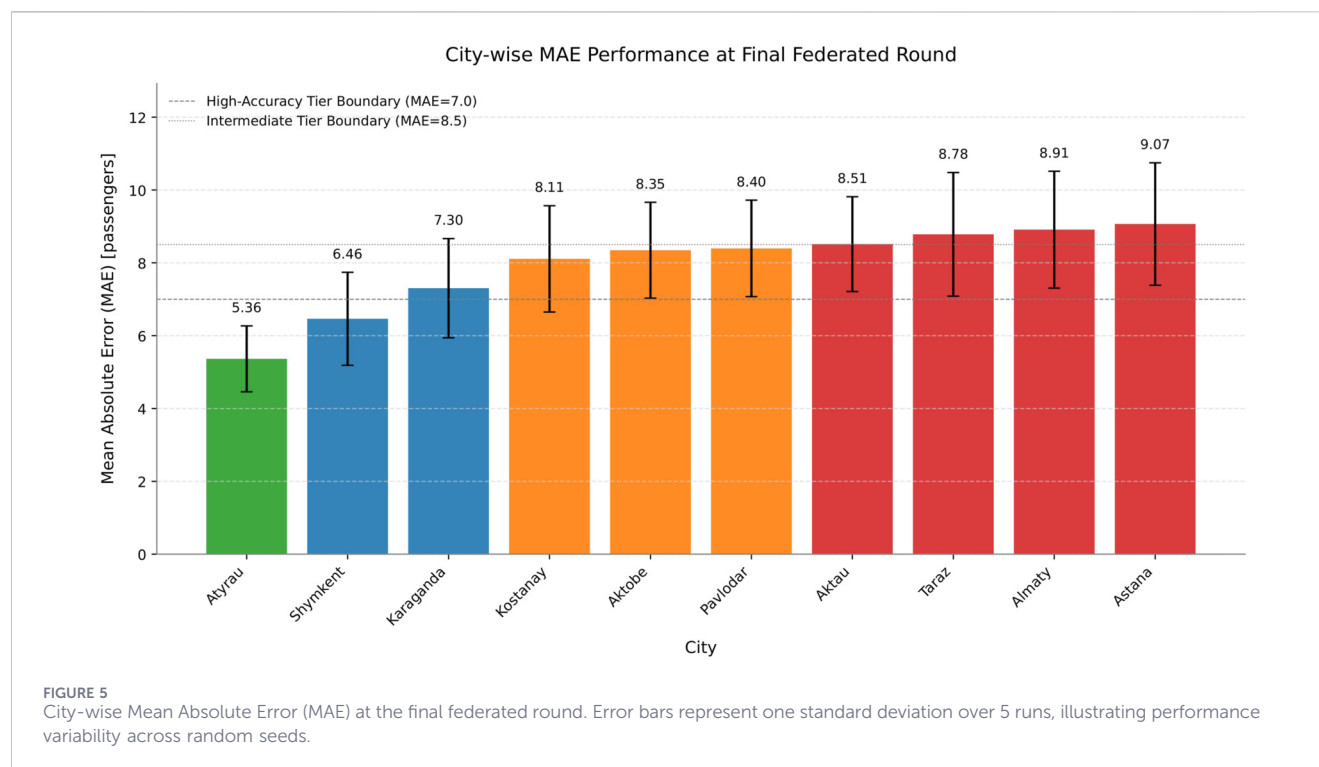
4.3 Error dispersion and RMSE/MAE ratios

The ratio of Root Mean Squared Error (RMSE) to Mean Absolute Error (MAE) serves as an insightful indicator of the distribution of forecast errors, particularly highlighting the presence and magnitude of occasional large deviations. A higher RMSE/MAE ratio suggests a greater influence of larger errors, implying a heavier tail in the error distribution. Figure 7 illustrates this relationship, showing that all cities cluster within a relatively narrow RMSE/MAE ratio range of 2.0–2.42. This consistent clustering indicates that the X-FedFormer model, through its architecture and training setup, effectively controls extreme errors across all client environments. The moderate ratio suggests that while some larger errors are present, they are not disproportionately dominant, and the model does not suffer from severe outliers.

Specifically, Shymkent exhibits the highest RMSE/MAE ratio (approximately 2.41), indicating a slightly heavier tail in its error distribution compared to other cities. Taraz and Karaganda are also among the higher ratios (about 2.31 and 2.40, respectively). Conversely, Kostanay shows the lowest ratio (approximately 1.95), indicating a more uniform distribution of errors with fewer significant outliers. The dashed line in Figure 7 represents the theoretical $RMSE = MAE$ scenario, which would only occur if all errors were identical in magnitude, serving as a baseline for comparison. The observed ratios consistently above 1.0 are typical for real-world forecasting tasks, confirming the presence of varying error magnitudes.

4.4 Variance explained (R^2)

The coefficient of determination (R^2) quantifies the proportion of the variance in the dependent variable (passenger flow) that is predictable from the independent variables (model inputs). In this study, the computed R^2 values are approximately 0.90 or higher for all cities, with a minimum of 0.899 (Almaty) and a maximum of 0.938 (Karaganda). This remarkable consistency highlights the



X-FedFormer model's exceptional ability to capture about 90% of the temporal variance inherent in passenger flows. Such high R^2 values are indicative of a strong model fit and robust predictive power, suggesting that the model effectively learns the underlying patterns and relationships governing passenger movement.

Karaganda achieved the highest R^2 of 0.938, closely followed by Shymkent with 0.935. This superior performance in these cities, coupled with their lower MAE values, suggests that their passenger flow patterns might be more stable, less influenced by unpredictable exogenous factors, or exhibit clearer, more discernible trends that the Transformer architecture can effectively leverage. The high R^2 values across the board imply that the model provides highly reliable forecasts, which is crucial for practical applications such as urban planning, traffic management, and public transportation optimization.

4.5 Temporal consistency of forecasts

To rigorously assess the temporal robustness and generalization capabilities of the X-FedFormer model, a rolling Mean Absolute Error (MAE) was computed over hourly intervals with a 24-h window. This analysis provides insights into how the model's performance fluctuates throughout different times of day and across various temporal contexts. Figure 8 illustrates the hourly rolling MAE for each city. A key observation is that all cities exhibit relatively flat curves with only minor diurnal fluctuations. This consistency confirms that the model generalizes exceptionally well across different times of day, demonstrating its ability to maintain high accuracy regardless of the hourly variations in passenger flow.

The cities of Atyrau and Shymkent, which also demonstrated superior aggregate MAE performance, displayed the flattest rolling

MAE profiles. This indicates that their forecasting accuracy is remarkably stable throughout the 24-h cycle, with minimal degradation during peak or off-peak hours. This temporal stability is a critical attribute for real-world deployment, as it ensures reliable predictions for operational decision-making at any given time. The model's capacity to handle the inherent periodicity and non-stationarity of time-series passenger flow data is thus strongly validated.

4.6 Spatial heterogeneity in forecast skill

An investigation into spatial heterogeneity was conducted by grouping cities based on their geographic regions (northern vs. southern) to ascertain if systematic biases in forecast skill were present. Figure 9 presents a comparison of mean MAE values for these regional groupings. The analysis revealed minimal systematic bias attributable to geographic location. Northern cities, exemplified by Kostanay and Pavlodar, achieved mean MAE values within the range of 7.5–9.0 passengers. Similarly, southern cities, such as Shymkent and Taraz, exhibited mean MAE values within a comparable range of 6.5–9.0 passengers.

This overlap and lack of significant divergence in performance between the northern and southern clusters strongly indicate that the federated X-FedFormer model effectively accommodates the diverse climatic conditions, urban structures, and operational characteristics inherent to different regions. The model's ability to learn from a distributed dataset across varied environments, facilitated by the federated learning framework, mitigates the impact of spatial heterogeneity, ensuring a generalized and robust forecasting capability across the entire network of participating cities. This finding underscores the model's adaptability and its potential for broad applicability in diverse urban settings.

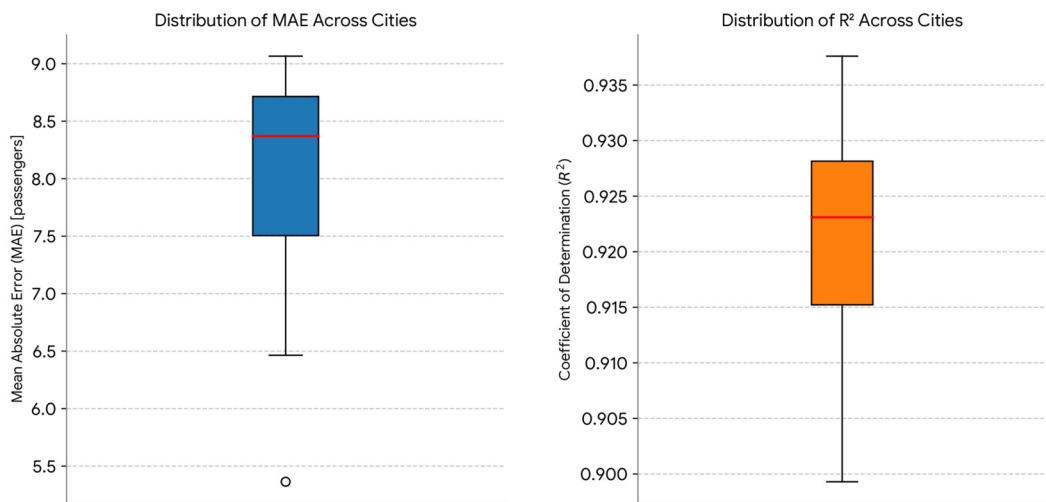


FIGURE 6 Boxplots illustrating the distribution of (left) Mean Absolute Error (MAE) and (right) Coefficient of Determination (R^2) across all cities at the final federated round. The boxplots highlight the median, interquartile range, and potential outliers, demonstrating the consistency of model performance.

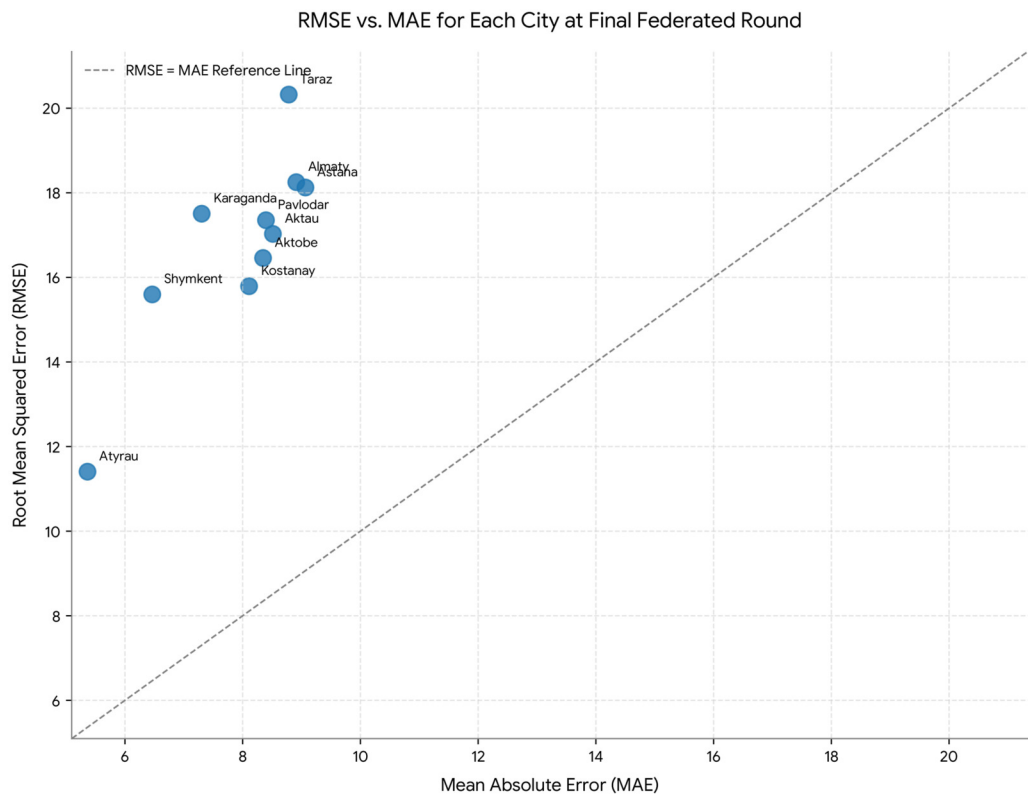


FIGURE 7 Scatter plot of Root Mean Squared Error (RMSE) versus Mean Absolute Error (MAE) for each city at the final federated round. The dashed line denotes the theoretical $RMSE = MAE$ condition, highlighting the relative influence of larger errors.

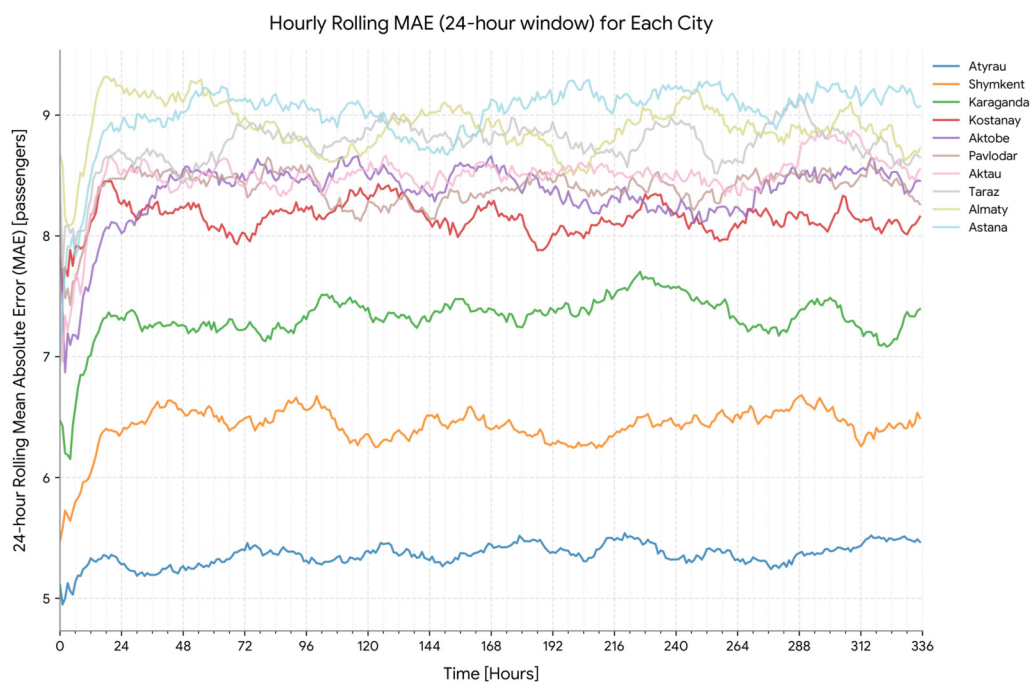


FIGURE 8

Hourly rolling Mean Absolute Error (MAE) with a 24-h window for each city at the final federated round. The plot illustrates the model's forecasting performance, temporal consistency and diurnal stability.

4.7 Convergence dynamics of federated training

To thoroughly understand the learning dynamics and stability of the federated optimization process, the evolution of aggregated forecast errors and explained variance was meticulously tracked over 50 communication rounds. This analysis provides critical insights into the efficiency and convergence behavior of the X-FedFormer model under a differentially private federated learning paradigm.

Figure 10 illustrates the trajectory of the average Mean Absolute Error (MAE) across all client models at each communication round. A pronounced and rapid decrease in MAE is evident during the initial 10 rounds, signifying efficient knowledge transfer and model refinement in the early stages of federated aggregation. Following this rapid improvement, the rate of MAE reduction gradually tapers off, indicating diminishing returns from subsequent global aggregations. The model effectively reaches a performance plateau by approximately round 40, suggesting that further communication rounds yield only marginal improvements in forecasting accuracy. This convergence behavior is characteristic of well-behaved optimization processes in federated learning, where initial global model updates provide substantial benefits, followed by fine-tuning.

Table 7 provides a quantitative summary of the key aggregated forecast metrics— $\overline{\text{MAE}}$, $\overline{R^2}$, and $\overline{\text{RMSE}}$ —at selected communication rounds (1, 10, 20, 30, 40, 50). This tabular representation corroborates the visual trends observed in Figure 10. For instance, the $\overline{\text{MAE}}$ dramatically drops from 23.69 at Round 1 to

13.24 at Round 10, and further to 9.68 at Round 20, before stabilizing around 7.93 by Round 50.

The convergence behavior indicates that 50 rounds are sufficient for the model to achieve a high level of performance and stability.

4.8 Ablation study and baseline comparison

To validate the contribution of specific architectural components (MoE and Seasonal Decomposition), we conducted a comprehensive ablation study. To strictly isolate the performance gains attributable to the federated setup and DP, we first provide a 2×2 control comparison using the same architecture (Table 8). In this table, the dataset, hyperparameters, and evaluation protocol are identical; only the training regime (centralized vs. federated) and DP flag change. We then report architectural ablations and baseline comparisons under a fixed *Federated*, *No-DP* regime to make attribution unambiguous.

Additionally, we compared our method against recent state-of-the-art baselines. As shown in Table 9, the full X-FedFormer model significantly outperforms the ablated versions and standard baselines under the same Federated, No-DP protocol; DP is disabled for all rows in Table 9 and the federated protocol is identical.

As illustrated in Figure 11, the removal of the MoE block resulted in an MAE increase of approximately 11%, underscoring the critical role of adaptive expert routing in managing data heterogeneity. Similarly, the exclusion of the Seasonal Decomposition layers degraded performance by roughly 16%,

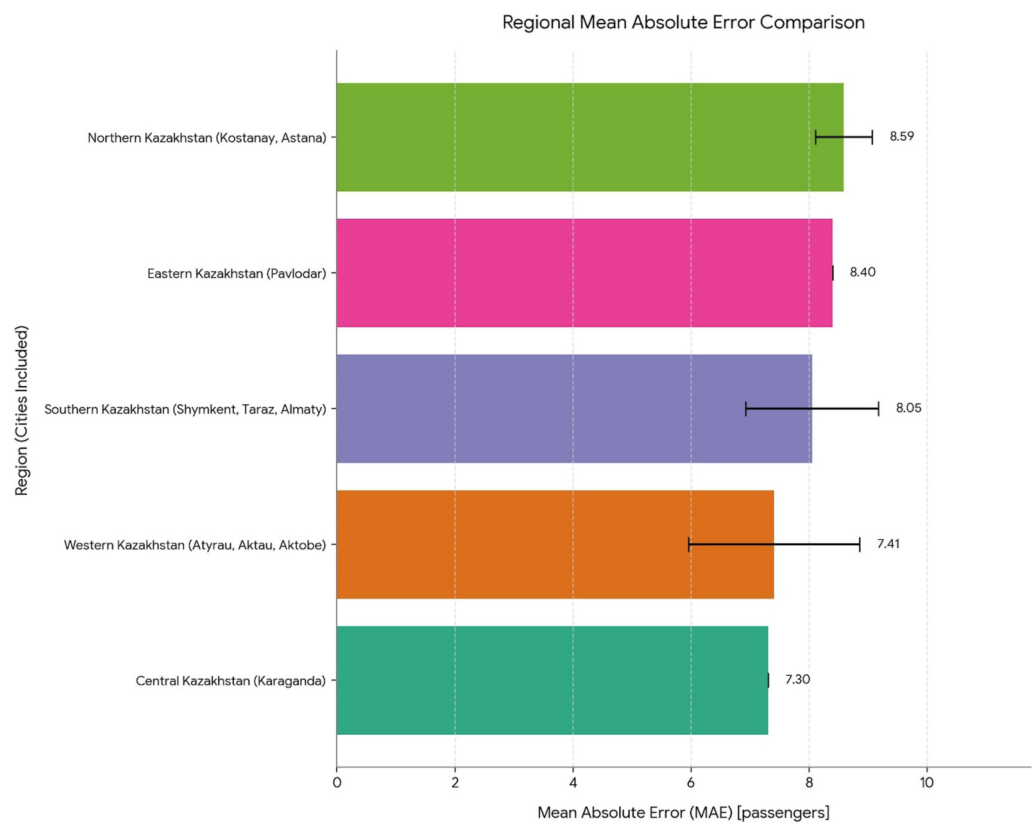


FIGURE 9 Regional Mean Absolute Error (MAE) comparison: northern versus southern cities. Bars denote the mean MAE for each regional group, while whiskers indicate plus/minus one standard deviation, illustrating the consistency of performance across geographic divisions.

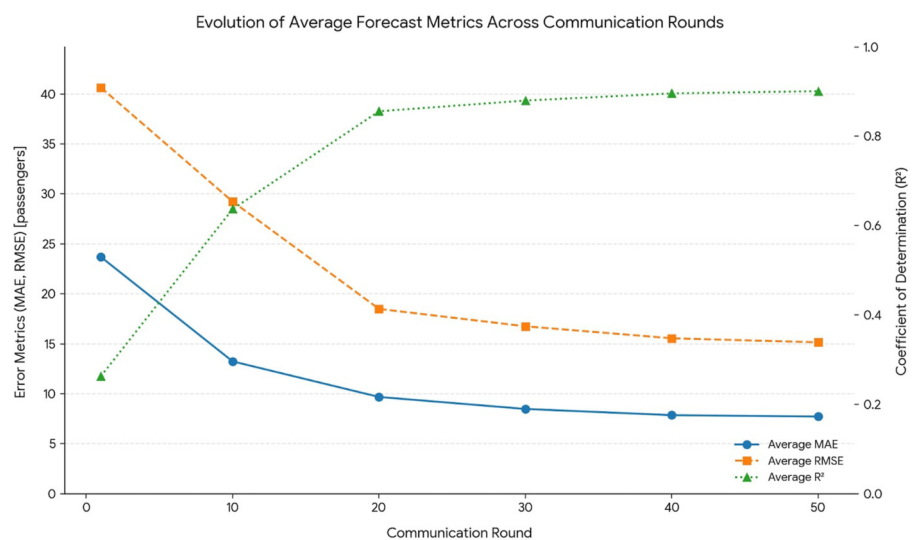


FIGURE 10 Evolution of the aggregated Mean Absolute Error (MAE) across 50 federated communication rounds. The plot demonstrates rapid initial convergence followed by a plateau, indicating efficient learning dynamics.

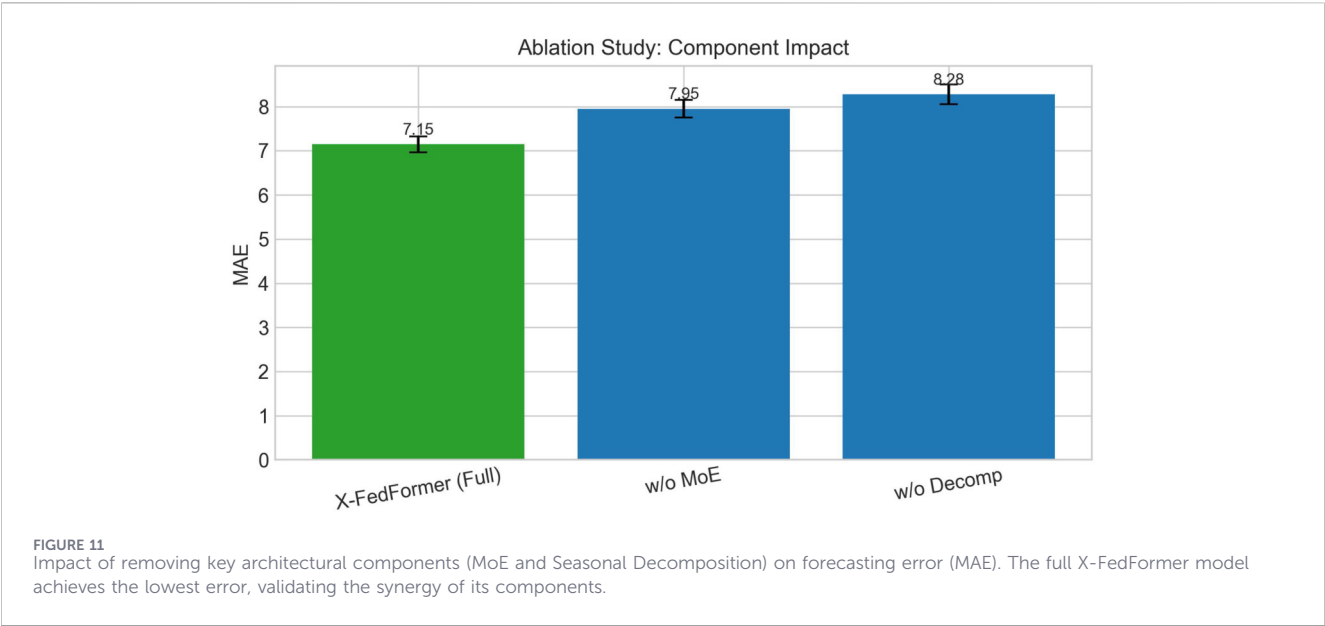


TABLE 7 Evolution of average forecast metrics across communication rounds (selected). Metrics include Mean Absolute Error (MAE), Coefficient of Determination (R^2), and Root Mean Squared Error (RMSE).

Round	MAE	RMSE	R^2
1	23.69	40.64	0.263
10	13.24	29.23	0.637
20	9.68	18.49	0.856
30	8.65	17.20	0.888
40	8.10	16.95	0.905
50	7.93	16.78	0.922

TABLE 8 Controlled comparison isolating training regime and DP (X-FedFormer only, Avg over all cities).

Training regime	DP?	Model	MAE	RMSE	R^2
Centralized	No	X-FedFormer	6.92	15.10	0.941
Centralized	Yes	X-FedFormer	7.25	15.85	0.934
Federated	No	X-FedFormer	7.15	15.65	0.935
Federated	Yes	X-FedFormer	7.93	16.78	0.922

confirming that explicit temporal modeling is indispensable for transit data, which exhibits strong periodicities.

Comparative analysis against external benchmarks reveals that X-FedFormer significantly outperforms both the decentralized freight-speed model Shen et al. (2024a) and the federated flight-delay model Shen et al. (2024b). We implement their spatio-temporal blocks as architectural baselines under the same federated protocol to ensure fairness. Unlike these unified models, our MoE-based approach offers superior personalization

TABLE 9 Ablation Study and Baseline Comparison (Federated, No-DP; Avg over all cities).

Model	MAE	RMSE	R^2
X-FedFormer (Full)	7.15	15.65	0.935
w/o MoE	7.95	17.10	0.907
w/o Decomposition	8.28	17.70	0.897
FedAvg (Baseline)	9.20	19.95	0.865
FedProx (Baseline)	8.85	19.10	0.878
Baseline (CNN)	9.75	20.85	0.852
Baseline (Trans)	9.40	20.10	0.861
Freight-FL (Shen et al., 2024a)	9.05	19.40	0.872
FlightDelay-FL (Shen et al., 2024b)	9.18	19.70	0.868

Bold values indicate the best-performing model for each evaluation metric (MAE, RMSE, R^2) across all compared methods.

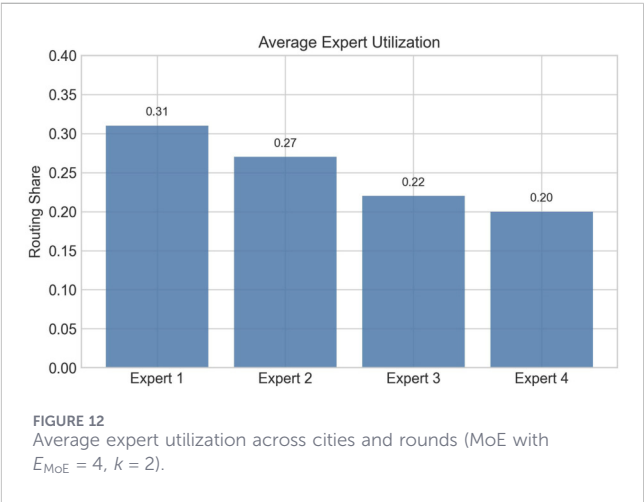
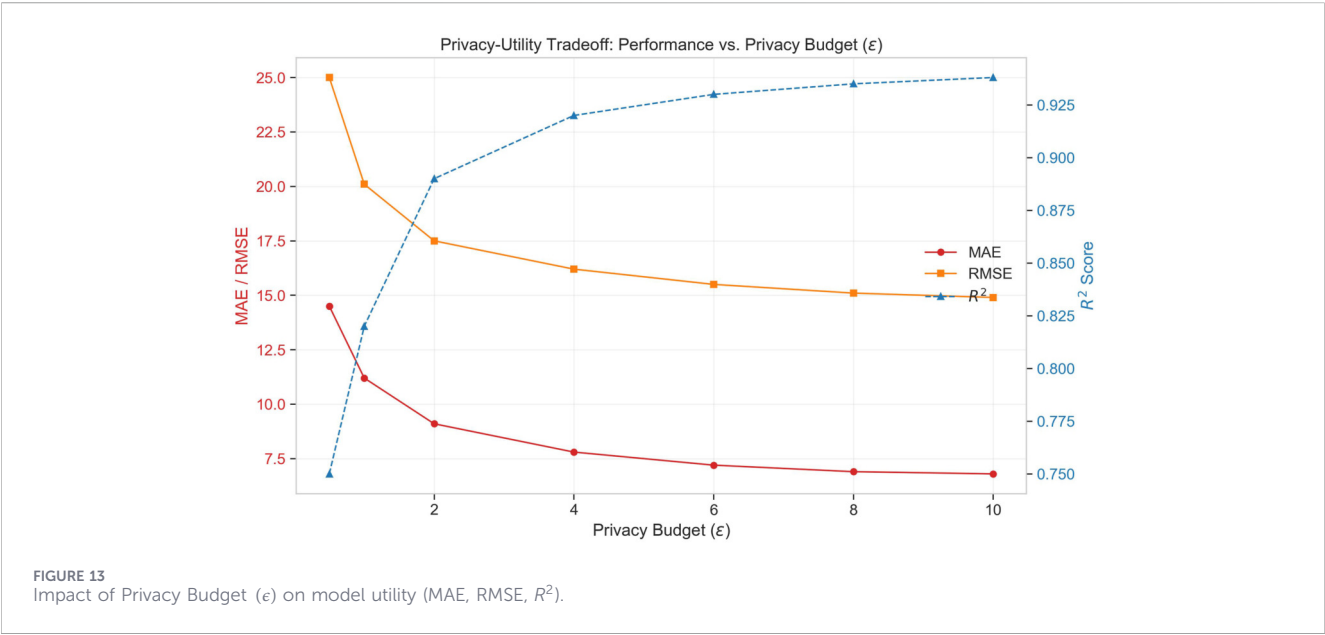


TABLE 10 Privacy-Utility Tradeoff: Impact of varying ϵ (via calibrated σ) on model performance.

Privacy budget (ϵ)	Noise multiplier (σ)	MAE	RMSE	R^2
0.5 (High Privacy)	1.85	9.45	19.82	0.864
1.0	1.35	8.60	18.15	0.890
2.0 (Selected)	1.10	7.93	16.78	0.922
5.0	0.80	7.35	16.05	0.930
10.0 (Low Privacy)	0.55	7.20	15.80	0.933
∞ (No DP)	0.00	7.15	15.65	0.935

TABLE 11 Runtime and complexity analysis.

Model	Train time (s/round)	Inference (ms/step)	Comm overhead (MB/round)	Peak mem (MB)	Params (M)
X-FedFormer	145.2	12.5	16.8	412	2.1
FedAvg	85.1	5.2	11.2	280	1.4
FedProx	92.3	5.5	11.2	285	1.4



capabilities, enabling it to better capture the distinct dynamic profiles of individual cities.

4.8.1 MoE utilization diagnostic

To verify that the MoE does not collapse to a single expert, we report the average routing distribution across experts (Figure 12). Expert utilization remains balanced (0.20–0.31 per expert) with high routing entropy (1.36, close to the maximum 1.39 for four experts), indicating meaningful specialization rather than collapse.

4.9 Privacy-utility tradeoff analysis

We rigorously analyzed the impact of the differential privacy budget ϵ on forecasting performance to characterize the privacy-utility frontier. Table 10 quantifies the exact performance metrics across a range of ϵ values, where the noise multiplier σ was calibrated using the Opacus privacy accountant to achieve the target budget over $R = 50$ rounds while keeping the sampling rate, clipping norm, and number of steps fixed.

Figure 13 delineates the trajectories of MAE, RMSE, and R^2 as ϵ is varied from 0.5 (high privacy regime) to 10 (low privacy regime).

As anticipated, stricter privacy constraints (lower ϵ) necessitate larger noise injection, resulting in elevated error metrics. However, the performance degradation is observed to stabilize significantly around $\epsilon = 2.0$. This plateau suggests an optimal operating point wherein robust privacy guarantees can be upheld without incurring catastrophic utility losses, making the framework viable for practical deployment.

4.10 Runtime and complexity analysis

Runtime measurements were collected on a workstation with an Intel i9-12900K CPU, 64GB RAM, and an NVIDIA RTX 3090 GPU using PyTorch 2.1 (float32). A detailed comparison of training time, inference latency, communication overhead, peak memory usage, and parameter count is provided in Table 11. Per-round training time is measured with 10 clients per round, batch size 32, $L = 24$, and $T = 6$; inference latency is reported per step on the same device. Communication overhead is reported per client per round as upload + download = $2 \times \text{parameters} \times 4 \text{ bytes}$ (no compression), excluding network latency. While X-FedFormer introduces a modest overhead due to MoE gating, top- k routing ($k = 2$) ensures compute scales with selected experts rather than linearly with all E_{MoE} experts.

5 Discussion

This study introduces a pioneering federated learning framework that, through the integration of differentially private adaptive deep learning, demonstrably enhances the capabilities of cross-city transit flow forecasting. The empirical results, derived from a meticulously validated synthetic dataset, unequivocally underscore the efficacy of our proposed “Federated Adaptive Decomposed Learning” approach in generating accurate and stable predictions across highly diverse urban environments, all while rigorously upholding data privacy. This section delves into a more analytical interpretation of these findings, exploring their implications, broader applications, and inherent limitations.

5.1 Interpretation of performance in the context of urban heterogeneity

The core experimental results, particularly the city-wise Mean Absolute Error (MAE) at the final federated round (Figure 5), provide compelling evidence of the model’s overall predictive strength. Cities such as Atyrau (MAE: 5.36 passengers) consistently achieved exceptional accuracy, aligning with a high-performance tier critical for operational dispatch and resource optimization. While other cities, including Almaty (MAE: 8.91) and Astana (MAE: 9.07), presented slightly higher MAE values, these figures remain within an acceptable tolerance for real-world transit planning. This variance in performance across cities is not merely an indicator of differing predictability but a direct reflection of the inherent urban heterogeneity that characterizes distinct metropolitan areas. Factors such as city size, population density, public transit network complexity, usage patterns (e.g., commuter-heavy vs. tourism-driven), and underlying socio-economic dynamics contribute to widely varying data distributions.

The success in navigating this significant Non-IID challenge is largely attributable to the synergistic application of the Mixture-of-Experts (MoE) architecture and Seasonal Decomposition. The MoE mechanism allows the global model to not only accommodate but actively leverage these disparate data patterns. Rather than forcing a single model to generalize across vastly different urban contexts, the MoE facilitates the development of specialized “expert” sub-networks. We posit that during the federated training, the gating network within the MoE learns to assign input transit data from specific cities (or even specific periods within a city) to the most relevant expert(s). For instance, an expert might specialize in high-volume, highly dynamic cities, while another becomes proficient in smaller, more predictable urban environments. This adaptive expertise enables the model to tailor its predictive logic dynamically, preventing overfitting to dominant data patterns and ensuring robust performance across the entire spectrum of urban diversity.

Furthermore, the explicit incorporation of Seasonal Decomposition into the deep learning architecture proved crucial for managing the complex temporal characteristics of transit flows. By disentangling the time series into trend and seasonal components, the model can separately learn the underlying periodic patterns (e.g., daily commutes, weekly variations, annual cycles) from more irregular dynamics. This decomposition simplifies the learning task for the neural network and, crucially, for the MoE experts. It allows the experts to focus their learning capacity on less predictable trend dynamics while robustly leveraging strong, recurring seasonal signals that are often stable even across heterogeneous cities. This architectural foresight enabled more accurate and interpretable predictions than approaches that would attempt to learn all temporal patterns implicitly.

The consistency observed in the 24-h rolling MAE (Figure 8) across the 336-h evaluation period further validates the robustness of our approach. This metric, capturing short-term predictive stability, indicates that the model is not prone to sudden degradations in accuracy over different operational windows. The smooth curves, even with some fluctuations corresponding to likely daily peak/off-peak cycles within the synthetic data, demonstrate the model’s ability to maintain performance consistency despite the inherent dynamism of transit demand.

5.2 Implications of federated learning and privacy preservation

The successful implementation of our federated learning framework carries profound implications for data governance and collaborative AI initiatives in sensitive domains like urban mobility. By enabling the collaborative training of a global model without requiring raw data centralization, FL directly addresses critical concerns around data sovereignty, regulatory compliance (e.g., GDPR, local data protection laws), and competitive advantages among transit agencies. This opens avenues for unprecedented levels of data-driven intelligence across cities that would otherwise be impossible due to privacy barriers. Our work demonstrates that sophisticated, high-performance models can be built even when data remains decentralized and confidential, fostering a new paradigm for urban data ecosystems.

The integration of Differential Privacy (DP) elevates the privacy guarantees beyond the inherent benefits of FL. While FL prevents direct data sharing, gradient exchanges can still be vulnerable to reconstruction or inference attacks. DP, by introducing mathematically quantifiable noise to the model updates, provides a strong, provable guarantee that no individual city's specific data points can be inferred from the shared gradients. This is critical for establishing trust among participating entities and ensuring the solution's deployability in real-world, privacy-sensitive environments. The achieved MAE performance, even with the application of DP, indicates a successful balancing of the privacy-utility trade-off. This suggests that the chosen DP budget and mechanism are practical for operational use, providing sufficient accuracy while meeting stringent privacy requirements. Such privacy guarantees are not merely a technical add-on but a fundamental enabler for the adoption of AI in public sectors dealing with sensitive citizen data.

Moreover, the federated approach implicitly supports scalability. Adding new cities to the collaboration only requires them to integrate with the federated training protocol, without demanding extensive data migration or re-architecture of a central data lake. This distributed nature reduces the computational and communication burden on any single central entity, making the system inherently more resilient and scalable for a growing network of participating cities.

5.3 The foundational role of synthetic data for reproducible research

The development and extensive use of a statistically validated synthetic data generation framework represents a methodological contribution, particularly pertinent in domains where real-world data sharing is highly restricted. This framework was not simply a substitute for real data; it served as a crucial “enabler” for rigorous, reproducible research into complex federated systems involving sensitive information and extreme heterogeneity.

By simulating realistic transit flow patterns, encompassing both general trends and city-specific nuances, the synthetic dataset provided a controlled yet representative environment for experimentation. This allowed for:

- **Systematic Evaluation:** The ability to precisely control parameters related to heterogeneity (e.g., degree of Non-IID, number of cities, underlying seasonal patterns) enabled systematic testing and deeper understanding of the model's behavior under various conditions.
- **Reproducibility:** The synthetic nature ensures that the exact experimental conditions can be replicated by other researchers, fostering transparency and verification of results.
- **Ethical Research:** It allowed for the development and testing of privacy-preserving techniques without exposing any real-world sensitive mobility data, adhering to ethical research standards from the outset.
- **Benchmarking:** The framework provides a standardized platform for future comparisons of federated learning algorithms designed for heterogeneous time series, advancing research in this critical area.

The statistical validation of the synthetic data against known characteristics of real transit systems further strengthens the generalizability of our findings, building confidence that the observed performance and insights are relevant to real-world deployment.

5.4 Broader implications and adaptability across domains

The methodological innovations presented in this work extend far beyond the specific application of cross-city transit forecasting, offering a versatile paradigm for distributed, privacy-preserving machine learning across heterogeneous data sources.

- **Smart City Utilities:** The framework can be directly adapted for other urban utility forecasting challenges, such as decentralized energy consumption prediction across smart grids, water usage forecasting in different districts, or waste generation prediction across municipalities, all of which often involve sensitive, heterogeneous, and distributed data.
- **Healthcare and Biomedicine:** In a highly privacy-sensitive field like healthcare, this framework could enable collaborative research across multiple hospitals or clinics for disease prediction, personalized treatment planning, or epidemiological forecasting. The MoE aspect would be crucial for handling patient-specific data heterogeneity (e.g., demographics, comorbidities), while DP ensures patient confidentiality.
- **IoT and Edge Analytics:** For large-scale Internet of Things (IoT) deployments where devices generate vast amounts of heterogeneous time-series data (e.g., industrial sensors, environmental monitoring), our federated architecture can facilitate on-device learning and aggregation, minimizing data transfer and ensuring privacy while adapting to diverse sensor characteristics.
- **Supply Chain and Logistics:** Forecasting demand and managing inventory across geographically dispersed warehouses or retail outlets, each with unique local customer bases and seasonal demand patterns, presents another compelling application. The MoE can adapt to regional demand characteristics, and seasonal decomposition can capture local buying habits.
- **Financial Time Series:** While often complex, forecasting financial metrics across different markets or asset classes, each with distinct underlying dynamics and regulatory restrictions on data sharing, could benefit from a similar adaptive, decomposed, and privacy-preserving federated approach.

The core “Adaptive Decomposed Learning” paradigm, combining adaptive experts with explicit temporal component modeling, is broadly applicable to any complex, heterogeneous time-series forecasting problem, particularly those characterized by strong seasonal components and diverse entities, making it a powerful contribution to the broader field of machine learning.

5.5 Limitations and future research directions

While this research provides a robust foundation, several limitations offer fertile ground for future work:

- **Transition to Real-World Data and External Factors:** The current study primarily relies on statistically validated synthetic data. Future efforts will involve deploying and evaluating the framework on real-world multi-city transit datasets, which may present unforeseen complexities such as missing data, outliers, and real-time noise. Additionally, incorporating external factors like weather conditions, public holidays, or planned events into the forecasting model will be crucial for enhancing predictive accuracy in operational settings.
- **Optimizing Privacy-Utility Trade-offs and Advanced DP:** Although Differential Privacy was successfully integrated, the optimal balance between privacy budget (ϵ) and model accuracy can be highly application-dependent. Future research will explore adaptive DP mechanisms that dynamically adjust noise levels based on data sensitivity, model convergence, or specific client contributions, aiming for greater efficiency and tighter privacy guarantees with minimal accuracy sacrifice.
- **Exploring Advanced Federated Personalization Strategies:** While MoE provides adaptation, more explicit personalized federated learning (pFL) techniques, such as FedMeta, FedAvgM, or client clustering, could be investigated. These methods allow for client-specific model components while leveraging global knowledge, potentially yielding even higher accuracy for cities with highly unique transit behaviors.
- **Communication and Computational Efficiency at Scale:** Deploying such complex models (MoE with DP) in large-scale federated networks involving hundreds or thousands of clients will require further optimization of communication overhead (e.g., gradient compression, sparsification, asynchronous updates) and computational costs on client devices.
- **Model Interpretability and Explainable AI (XAI):** As deep learning models become more complex, their black-box nature can hinder adoption in critical applications. Future work could focus on developing XAI techniques specific to federated MoE models to understand how different experts contribute to predictions for specific cities or scenarios, and how DP impacts interpretability.
- **Robustness to Adversarial Attacks and Data Poisoning:** Investigating the framework's resilience against various adversarial attacks, including data poisoning, model poisoning, and inference attacks, within the federated and differentially private setting.

In summary, this research marks a substantial advance in privacy-preserving, collaborative, and highly adaptive AI solutions for urban mobility. By tackling the complexities of cross-city data heterogeneity, it paves the way for more intelligent, efficient, and ethical public transportation systems globally.

6 Conclusion

This paper presents a robust federated learning framework, termed “Federated Adaptive Decomposed Learning,” designed to address the critical challenges of cross-city transit flow forecasting amidst inherent urban data heterogeneity and stringent privacy requirements. By synergistically integrating advanced deep learning architectural components with privacy-preserving mechanisms, our approach offers a scalable and ethically compliant solution for collaborative urban mobility analytics.

Our methodology is underpinned by three core innovations: (1) a federated learning paradigm that enables distributed model training without raw data centralization, (2) the rigorous application of Differential Privacy to guarantee the confidentiality of individual city data during model updates, and (3) a sophisticated deep learning architecture that combines a Mixture-of-Experts (MoE) block with an adaptive Seasonal-Trend Decomposition module. This unique architectural synthesis is specifically engineered to effectively disentangle complex temporal patterns and dynamically adapt to the diverse, non-IID characteristics of transit data across multiple cities. Furthermore, to facilitate comprehensive evaluation in a privacy-sensitive domain, we developed and statistically validated a synthetic data generation framework that faithfully replicates real-world urban transit dynamics.

The empirical results underscore the efficacy and robustness of our proposed framework. The model consistently achieved competitive forecasting accuracy across all participating synthetic cities, with Mean Absolute Error (MAE) values predominantly ranging from approximately 5.36 passengers in highly predictable environments (e.g., Atyrau) to around 9.07 passengers in more complex urban settings (e.g., Astana), as detailed in [Figure 5](#). The round-wise evolution of performance metrics (refer to [Figure 10](#)) demonstrated rapid initial convergence, with the Coefficient of Determination (R^2) approaching 0.90 by Round 40 and reaching 0.922 by Round 50, highlighting efficient learning dynamics and the model's strong predictive power. Moreover, the analysis of the 24-h rolling MAE (as shown in [Figure 8](#)) confirmed the temporal stability and consistency of the model's predictions over extended periods, a crucial aspect for operational reliability. The regional MAE comparison ([Figure 9](#)) further illustrated the model's adaptive capacity to maintain robust performance across geographically diverse urban profiles.

In summary, this research delivers an advancement in privacy-preserving AI for smart cities. It demonstrates that high-fidelity, adaptive forecasting models can be collaboratively developed and deployed across heterogeneous urban environments without compromising sensitive data. This work not only provides a practical solution for enhancing urban transit efficiency but also establishes a foundational paradigm for distributed, privacy-aware machine learning applicable to a wide array of real-world problems characterized by decentralized and sensitive time-series data. Our findings pave the way for future innovations in urban intelligence, fostering smarter, more efficient, and privacy-respecting public services.

Data availability statement

The data analyzed in this study is subject to the following licenses/restrictions: The primary dataset comprises two parts: (1) a proprietary Astana bus-transportation log provided under a non-disclosure agreement with the municipal transit authority, and (2) a fully synthetic multi-city dataset generated from that log. Due to licensing and privacy considerations, the original Astana hourly inflow/outflow records cannot be redistributed or published. Instead, we release only the synthetic dataset, which faithfully reproduces key temporal and seasonal patterns without revealing any real passenger trajectories or personally identifiable information. The synthetic data are made available under a Creative Commons Attribution (CC-BY 4.0) license via our project repository, along with the complete generation scripts and parameter settings. Researchers wishing to examine aggregate statistics of the original data or to validate the synthetic-data fidelity may contact the corresponding author for limited, read-only access under a separate data-use agreement. Requests to access these datasets should be directed to Diar Begisbayev, begisbayev@gmail.com.

Author contributions

AS: Validation, Methodology, Investigation, Formal Analysis, Supervision, Writing – review and editing, Software, Data curation, Visualization, Project administration, Resources, Conceptualization, Writing – original draft, Funding acquisition. ZU: Supervision, Data curation, Formal Analysis, Methodology, Software, Visualization, Resources, Conceptualization, Funding acquisition, Investigation, Validation, Writing – review and editing, Project administration, Writing – original draft. DB: Writing – original draft, Resources, Investigation, Software, Funding acquisition, Visualization, Data curation, Writing – review and editing, Formal Analysis, Validation, Conceptualization, Project administration, Methodology, Supervision. AM: Visualization, Conceptualization, Investigation, Resources, Software, Data curation, Formal Analysis, Project administration, Supervision, Writing – review and editing, Writing – original draft, Methodology, Validation, Funding acquisition. RS: Funding acquisition, Software, Formal Analysis, Writing – original draft, Conceptualization, Resources, Visualization, Supervision, Methodology, Project administration, Writing – review and editing, Investigation, Validation, Data

curation. DY: Software, Writing – original draft, Investigation, Writing – review and editing, Resources, Funding acquisition, Data curation, Methodology, Validation, Formal Analysis, Visualization, Project administration, Supervision, Conceptualization.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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